ATMega644 chip and stuff

X2
20MHz
x1
x2
C18
33pF
C19
33pF
dgnd

U9
ATMEGA644PA-AU
copi
cipo
clk
atmegaReset
+5V
dgnd
x2
x1
J1
tx0
D4
D5
D6
D7
res
tcq
PB5(PCINT13/ICP3/MOSI)
PB6(PCINT14/OC3A/MISO)
PB7(PCINT15/OC3B/SCK)
#RESET
VCC
GND
XTAL2
XTAL1
PD0(PCINT24/RXD0/T3)
PD1(PCINT25/TXD0)
PD2(PCINT26/RXD1/INT0)
PD3(PCINT27/TXD1/INT1)
PD4(PCINT28/XCK1/OC1B)
PD5(PCINT29/OC1A)
PD6(PCINT30/OC2B/ICP)
PD7(PCINT31/OC2A)
PC0(PCINT16/SCL)
PC1(PCINT17/SDA)
PC2(PCINT18/TCK)
PC3(PCINT19/TMS)
PA7(ADC7/PCINT7)
PA6(ADC6/PCINT6)
PA5(ADC5/PCINT5)
PA4(ADC4/PCINT4)
PA3(ADC3/PCINT3)
PA2(ADC2/PCINT2)
PA1(ADC1/PCINT1)
PA0(ADC0/PCINT0)
PB4(#SS/OC0B/PCINT12)
PB3(AIN1/OC0A/PCINT11)
PB2(AIN0/INT2/PCINT10)
PB1(T1/CLKO/PCINT9)
PB0(XCK0/T0/PCINT8)
AREF
ASCR
PC7(TOSC2/PCINT23)
PC6(TOSC1/PCINT22)
PC5(TDI/PCINT21)
PC4(TDO/PCINT20)

AtMega program header
R6
10kΩ
cipo
clk
atmegaReset
copi
dgnd

MB87186 and clock
SN74LVC2G14DB(LX)
3.58MHz
X1
R7
1MΩ
C21
33pF
C20
33pF
A5V
pin18
U10

U11
Casio FZ chip
D0
D1
D2
D3
D4
D5
D6
D7
rw
ce
chs
a
cq
R/W
CE
CHS
A
CQ
A5V
AudioIn1
A5V
AudioOut1
AudioOut2
A5V
AudioIn2
minus5
res
pin18
GND

Audio in
audio in 1
J5.1
R22
100kΩ
C24
1uF
U6.1
R23
100kΩ
R26
10kΩ
AudioIn1
R28
10kΩ
audio in 2
J6.1
R24
100kΩ
C25
1uF
U6.2
R25
100kΩ
R27
10kΩ
AudioIn2
R29
10kΩ

Audio outputs
AudioOut1
C22
22uF
R11
33kΩ
R10
10kΩ
R12
33kΩ
mix1
R13
1kΩ
U4.1
J2.1
audio out 1
AudioOut2
C23
22uF
R14
33kΩ
R15
10kΩ
R16
33kΩ
mix2
R17
1kΩ
U4.2
J3.1
audio out 2
mix1
mix2
R18
33kΩ
R19
33kΩ
R20
22kΩ
R21
1kΩ
U5.1
J4.1
audio out mix

Jack LEDs
VCC
J2.2
J3.2
J4.2
J5.2
J6.2
J7.2
J8.2
J11.2
J12.2
J9.2
J10.2
audio out 1
audio out 2
audio out mix
audio in 1
audio in 2
CV1IN
CV2IN
res1
res2
dca1
dca2
blue
red
green

Controls, CV and knob
A5V
res1
R50
minus5
J11.1
R51
49.9kΩ
R52
49.9kΩ
R48
49.9kΩ
R49
res1 atten
U2.1
R53
10kΩ
ADCRes1
R86
220Ω
D11
100kΩ
R82
U13.1
res2
R57
minus5
J12.1
R56
49.9kΩ
R55
49.9kΩ
R58
49.9kΩ
R59
res2 atten
U2.3
R54
10kΩ
ADCRes2
R87
220Ω
D12
100kΩ
R83
U13.2

DCFilter and Res inputs
dca1
J9.1
R60
dca1
R62
49.9kΩ
R61
49.9kΩ
R63
49.9kΩ
R64
10kΩ
ADCDCA1
U2.4
R88
220Ω
D9
100kΩ
R84
U13.4
dca2
J10.1
R69
dca2
R67
49.9kΩ
R68
49.9kΩ
R66
49.9kΩ
R65
10kΩ
ADCDCA2
U2.2
R89
220Ω
D10
100kΩ
R85
U13.3

DCFilter inputs
VCC
R35
1kΩ
R30
filterLeft0
R33
100kΩ
R34
100kΩ
R37
49.9kΩ
R36
100kΩ
R38
10kΩ
D5
ADCFilter1
D6
U3.2
U5.2
FilterLeft0CV
R31
CV1IN
R32
100kΩ
J7.1
R46
1kΩ
R45
filterRight0
R47
100kΩ
R42
100kΩ
R40
49.9kΩ
R41
100kΩ
R39
10kΩ
D7
ADCFilter2
D8
U7.1
U7.2
FilterRight0CV
R43
CV2IN
R44
100kΩ
J8.1

Regulated power
U8
78L05G-AB3-R
R5
47Ω
R81
47Ω
C5
22uF
C4
22uF
C2
100nF
C16
100nF
C17
100nF
R3
47Ω
R1
47Ω
R4
47Ω
R2
47Ω
R90
47Ω
U12
78L05G-AB3-R
C26
22uF
C27
22uF
C29
100nF
U2.5
R8
22kΩ
R9
22kΩ
U3.1
minus5

Power
U1
D4
D3
D2
D1
C1
22uF
C6
100nF
C8
100nF
C10
100nF
C12
100nF
C14
100nF
C3
22uF
C7
100nF
C9
100nF
C11
100nF
C13
100nF
C15
100nF
U7.3
U13.5

Derived from ALM/BusyCircuits Fizzle Guts project
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TITLE: FZilter	REV: 1.0
EasyEDA	Company: Modular for the Masses / Sheet: 1/1
	Date: 2026-04-10 Drawn By: juanito moore